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# CSE471: System Analysis and Design
# Assignment on Functional Requirements
# Proposed Project Title: Chhaya – A Personalized Teaching Style Learning Platform


| Group No: 05, CSE471 Lab Section: 14, Summer 2026 | |
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## Tech Stack:

● Language: Python, JavaScript, HTML, CSS
● Framework: FastAPI, React
● Styling: CSS3 (or Tailwind CSS)
● Database: Postgresql
● ORM: -
● Deployment: Vercel, Render
● API: Gemini, Google Calendar, Youtube-transcript-api, Edge TTS, Resend API/Firebase Cloud Messaging, Tesseract/Google Cloud Vision API

## Functional requirements:

**Module 1**

1. [MD.Mahidad Hossain] Students will be able to add and manage reference sources (such as YouTube playlists or course links). The system will automatically retrieve and clean the transcript (using youtube-transcript-api), analyze the instructor's teaching style (using Gemini), and generate a structured teaching-style profile (including pacing, vocabulary level, use of analogies, example density, and concept sequencing). Both the processed transcript and reusable teaching-style profile will be stored, allowing all subsequent AI generated explanations, summaries, quizzes, and study materials to adapt to the teaching style of the selected reference source.
2. [Lamia Salsabil Mim] Students will be able to generate AI-powered study guides for selected topics in their preferred teacher's teaching style. The system will also allow students to export the guide as a PDF, generate it in multiple languages, and produce a condensed formula sheet for STEM-related topics.
3. [Omar Faruk] Students will be able to upload and view scanned versions of past exam papers. The system will extract the raw text (using Tesseract/Google Cloud Vision API), clean up the raw text (using regex/spellcheck for local, cheap cleanup and using Gemini API for smart cleanup) and store the JSON file in the database. Which will become the reference for exam format and scope.
4. [Amiyo Bhowmik] Students will be able to monitor their learning activity through an interactive analytics dashboard. The system will track and aggregate study sessions, study guide views, learning path progress (when available), and overall platform engagement

over time, presenting visual charts and activity summaries that help students understand their usage patterns and learning history.

**Module 2**

1. [MD.Mahidad Hossain] Students will be able to save, view, edit, delete, and organize AI-generated teacher style profiles in a personal library. They will also be able to mark frequently used teacher profiles as favorites (or pin them), enabling quick access and allowing them to easily select their preferred teaching style for future AI-generated learning content.
2. [Lamia Salsabil Mim] Students will be able to learn topics that are not covered in their selected reference source. The system will generate new explanations in the selected teacher's teaching style and provide optional AI voice narration (using Edge TTS api)..
3. [Omar Faruk] Students will be able to generate likely exam questions based on previous exam papers. The system will analyze the OCR'd text (using Gemini API) to understand question types, point distribution, section breakdown, and typical phrasing style. Then the system will predict (using Gemini API) the likely exam questions.
4. [Amiyo Bhowmik] Students will be able to schedule periodic reviews of previously studied topics using a spaced repetition algorithm. Based on study history and learning activity tracked from Module 1, the system will automatically calculate optimal future review dates using a spaced repetition formula, maintain a structured review schedule, and notify students of upcoming revision sessions through the application's reminder system (Google Calendar API for event scheduling, Resend API / Firebase Cloud Messaging (optional)).

**Module 3**

1. [MD.Mahidad Hossain] Students will be able to request alternative analogies for any AI-generated explanation. The system will generate new analogies based on different real-life contexts (such as sports, cooking, gaming, or everyday situations) while preserving the selected teacher's teaching style, ensuring that explanations remain consistent in tone, vocabulary, and presentation.
2. [MD.Mahidad Hossain] Students will be able to indicate when they do not understand an explanation by selecting an option such as "I still don't get it." The system will automatically generate a revised explanation with simpler vocabulary, shorter sentences, and a slower, more gradual progression of concepts while maintaining the selected teacher's teaching style, ensuring the explanation is easier to understand without changing its overall tone and presentation.
3. [Lamia Salsabil Mim] Students will be able to generate interactive concept maps for selected topics. The system will create a hierarchical visual representation of related

concepts, helping students understand topic structure and the relationships between key ideas.

4. [Lamia Salsabil Mim] Students will be able to generate personalized learning paths that organize topics into a structured sequence of lessons. They will also be able to bookmark individual lessons for quick access and future revision.
5. [Omar Faruk] Students will be able to give quizzes similar to past year's questions. The system will generate new practice questions (using Gemini API) matching the detected exam profile's format and not a generic question bank.
6. [Omar Faruk] Students will be able to evaluate their quizzes and get a grade. The system will grade a student's typed/short answers (using Gemini API) against a rubric inferred from patterns in the uploaded paper's structure (not simple keyword matching). The system will send the grade to the student's email (using Resend API).
7. [Amiyo Bhowmik] Students will be able to view their weak learning areas based on quiz and mock-exam performance. The system will automatically analyze quiz results stored in the database, calculate topic-wise performance, and identify topics that consistently fall below a predefined performance threshold (via Gemini API). These weak topics will be displayed on the student's dashboard and used to personalize future study recommendations.
8. [Amiyo Bhowmik] Students will be able to receive personalized study recommendations based on their identified weak topics and learning progress (Youtube Data API for video suggestion). The system will prioritize topics that require additional practice and use the Gemini API to generate a customized study plan, including the recommended study order, revision tips, and estimated study time for each topic. The generated study plan will be saved so students can revisit previous recommendations.